Index research app

What's built, how to run it, and where every piece lives. Review copy, Nov 6, 2026. Market data runs through the Nov 5 close.

> **Where it all is:** everything is in the folder the app made for this session. Every path below starts from that folder (for example `output/report.html`). With the helper running, the hub opens at **http://localhost:8765**. Deleting this session also deletes that folder.

At a glance

One **hub** in your browser with four tabs. Each tab is its own page, and they never mix.

Tab	What it answers	Data it uses	How it updates
Indexes	How SPY, QQQ and friends have normally behaved: history, base rates, risk	Yahoo prices, Cboe VIX family, FRED yields, the events file	Automatically, when you run it
Tickers	Each of the 9 ETFs on its own page: stats, charts, order-flow stand-ins, news	The Indexes tab's daily table, Yahoo headlines, the Fed's press feed	Automatically, when you run it
Leverage (perps)	What each Kalshi perp offers, and what leverage costs and risks	Kalshi's public API, plus the Indexes tab's price history	Automatically, when you run it
GEX / DIX	What dealer positioning says about the next session's size	A SqueezeMetrics CSV you download, plus the other two tabs	Only when you click Update

Data flows one way. The Tickers and Leverage tabs read the Indexes tab's history, and the GEX / DIX tab reads Indexes and Leverage. Nothing reads the GEX / DIX tab, so an old GEX copy never affects the other tabs. **Only the GEX / DIX tab has a temperature (Cold / Warm / Hot).** Everything else is data plus fixed rules.

Inside the app

	Indexes	Tickers	Leverage	GEX / DIX
Pages	1	10 (overview + 9 tickers)	1	1
Studies	7	6-7 per ticker	4 + a sortable market table	3
Luck tests	30 (about 1.5 would pass by chance)	None (descriptive)	None (facts and estimates)	3
Passed (possible edges)	8: 4 direction, 4 size	-	-	1 (size)
Charts	11	6 per ticker	2	3

1. How to run it

One-time setup (already done in this folder):

```
.venv/Scripts/python -m pip install -r requirements.txt
```

Refresh everything (a minute or two). This downloads fresh data for Indexes, Leverage and Tickers (including new headlines), then rebuilds GEX / DIX from your saved copy and resets its temperature:

```
.venv/Scripts/python run_all.py
```

Open the hub. Start the helper, leave it running, then open http://localhost:8765. It only listens on this computer; press Ctrl+C to stop it.

```
.venv/Scripts/python hub_server.py
```

Command	What it does
`run_all.py --offline`	Rebuild every tab from the last downloads (no internet)
`run_stage1.py`	Indexes tab only. Add --open to open it, or --offline to skip downloading.
`run_stage1.py --check`	Self-check: no look-ahead, arithmetic, labels. Prints PASS or FAIL and writes no report.
`run_tickers.py`	Tickers tab only (add --offline to reuse the saved headlines)
`run_leverage.py`	Leverage tab only
`run_gex.py --import`	Copy the newest DIX CSV from Downloads and rebuild the GEX / DIX tab (same as the Update button)

A suggested routine until stage 2 automates it

- **After 4:30 pm ET** (the app skips the current day until then): run `run_all.py`.
- **Start the helper and open the hub.** Look at the badge on the **GEX / DIX** tab name.
- **If it says Hot or Warm without a check mark:** on the GEX / DIX tab, click **Get the file**, click **CSV** on SqueezeMetrics' page, then click **Update**.
- **Read the tabs.** There's no daily up/down call yet; that's stage 2.

2. Where everything is

All paths start from the folder the app made for this session.

Pages you look at

What	Address (with the helper running)	File
Hub (opens the Indexes tab)	http://localhost:8765	`output/index.html`
Indexes tab	http://localhost:8765/report.html	`output/report.html`
Tickers tab (overview)	http://localhost:8765/tickers.html	`output/tickers.html`
One ticker, e.g. SPY	http://localhost:8765/ticker_SPY.html	`output/ticker_SPY.html` (also QQQ, IWM, DIA, RSP, TQQQ, SQQQ, UPRO, SPXU)
Leverage tab	http://localhost:8765/leverage.html	`output/leverage.html`
GEX / DIX tab	http://localhost:8765/gex.html	`output/gex.html`

Commands (the top of the folder)

File	Role
`run_all.py`	Runs all four tabs in order: Indexes, Leverage, Tickers, then GEX / DIX
`run_tickers.py`	Builds the Tickers tab (overview + 9 ticker pages)
`run_stage1.py`	Builds the Indexes tab
`run_leverage.py`	Builds the Leverage tab
`run_gex.py`	Builds the GEX / DIX tab (--import copies a new file in first)
`hub_server.py`	The helper: serves the hub and runs the Update button

Code

Folder / file	What's inside
`stage1/sources.py`	Downloads: Yahoo ETFs (2010 on, plus SPY/QQQ long history), Cboe VIX family, FRED yields. Caches everything.
`stage1/dataset.py`	Builds the daily table (each column tagged with when it was known) and the long-history table
`stage1/studies.py`	The luck test and verdict rules, plus the 7 studies: study_overnight, study_vix, study_day_after, study_together, study_events, study_leveraged, study_risk
`stage1/health.py`	The Indexes tab's data check
`stage1/selfcheck.py`	The --check command
`stage1/report.py, report_template.html`	The page builder and template shared by all three tabs (charts, tab bar, dark mode, the GEX badge)
`tickers/news.py`	Headlines: Yahoo per ticker, the Fed's press-release RSS, and the upcoming-events calendar
`tickers/build.py`	Per-ticker stats: snapshot, order-flow stand-ins, overnight vs daytime, after each SPY day, events, risk, 3x tracking
`tickers/report.py`	The Tickers overview and the 9 ticker pages, with the ticker-button row
`leverage/kalshi.py`	Kalshi downloads: market list, risk settings, funding history, hourly prices. Also appends the snapshot log.
`leverage/build.py`	One row per perp, the 4 index-perp studies, and the data check
`leverage/report.py`	The Leverage page, with the sortable and filterable table
`gex/data.py`	Loads your GEX/DIX copy, lines it up with the Indexes data, and holds the Cold / Warm / Hot rules
`gex/importer.py`	Finds the newest DIX*.csv in Downloads, checks it, and copies it in
`gex/build.py, gex/report.py`	The GEX / DIX tab's reading, charts, history tests, data check and page

Data

File	What it is	Who updates it
`data/events/events.csv`	Jobs, CPI and Fed dates, 2010 through Dec 2027	You, by hand, when BLS posts 2028 dates or moves one
`data/manual/DIX.csv`	Your GEX/DIX copy	You, through the Update button
`data/kalshi/snapshots.csv`	A Kalshi market snapshot from every run (spreads and open interest Kalshi doesn't keep)	The app, each Leverage run
`data/raw/`	Cached downloads, including headlines in data/raw/news/ (safe to delete; re-downloaded next run)	The app

Spreadsheets the app writes (output/)

File	What it is
`daily_table.csv`	4,236 trading days (2010 to Nov 5, 2026) by 110 columns
`data_dictionary.csv`	What each column means, and when you could have known it
`long_history.csv`	SPY since 1993 and QQQ since 1999, with VIX (8,500 days; risk numbers only)
`leverage_markets.csv`	The Leverage tab's market table (88 perps)
`gex_daily.csv`	The daily table plus GEX/DIX, as known before each open
`gex_status.json`	The current Cold / Warm / Hot level (read by the tab badge)

Docs to review

File	What's in it
`APP_GUIDE.md`	The full guide: every tab, how to run it, and the Cold / Warm / Hot rules
`STAGE1_REVIEW.md`	Every Indexes-tab result, with tables
`BUILD_LOG.md`	How each piece was built, the fixes along the way, and how it was checked
`README.md`	Setup and run steps
`requirements.txt`	Exact package versions

3. Indexes tab

Data: 9 ETFs since 2010 (SPY, QQQ, IWM, DIA, RSP, TQQQ, SQQQ, UPRO, SPXU), SPY since 1993 and QQQ since 1999 for risk numbers, the VIX family, yields, and the event dates. **Where:** `output/report.html`; code in `stage1/`.

The rules every test follows

- **Day types** compare SPY's move with the move VIX expected (VIX / 16). Big drop is below -1; normal drop -1 to -0.3; quiet within 0.3; normal rise 0.3 to 1; big rise above +1.
- **Luck test:** 4,000 random picks of the same number of days (40,000 for borderline results). 5-day results slide the signal days along the calendar instead, because their windows overlap.
- **Possible edge** = luck under 5%, the same direction in at least 3 of 4 time periods, and 30 or more cases. **Weak hint** = luck under 20%. Events are compared with the same weekday.

The seven studies

#	Study	Headline result
1	Overnight vs. daytime	68% of SPY's growth since 2010 came overnight (QQQ 69%)
2	Fear gauge vs. move size	VIX overstates moves by about 15%; VIX9D is the best size predictor; the fear-curve shape doesn't predict direction
3	The day after (next day, overnight/daytime split, next 5 days)	Big rise: next day -0.12%, and -0.32% when VIX is 20 or higher (mostly overnight). Normal drop: next 5 days +0.51%. Big rise: next 5 days -0.02%.
4	Moving together	SPY, QQQ, DIA and RSP are basically one bet; QQQ moves about 1.13x SPY
5	Event days	Jobs mornings gap 1.19x and Fed afternoons swing 1.29x; no event predicts direction
6	3x ETFs	In choppy months TQQQ and SQQQ are each about 1.2% short of exact 3x, and both lose in 21% of them
7	How far it can go against you (since 1993)	A 1-in-100 day is 1.4% at VIX under 13 and 8.9% at VIX 30+. A 2-expected-move stop gets hit 1 day in 40. Weekend gaps 1.14x, holiday gaps 1.50x.

What passed (8 possible edges out of 30 tests)

Type	Result	Normal	Luck	Periods
Direction	After a big rise, next day -0.12%	+0.06%	0.9%	4 of 4
Direction	After a big rise with VIX 20 or higher, next day -0.32% (112 cases; split picked after looking)	+0.06%	3.5%	3 of 4
Direction	After a normal drop, next 5 days +0.51%	+0.28%	1.2%	4 of 4
Direction	After a big rise, next 5 days -0.02%	+0.28%	3.0%	4 of 4
Size	Jobs-report mornings: overnight gap 1.19x ordinary	1.00x	3.1%	3 of 4
Size	Fed afternoons: daytime move 1.29x ordinary	1.00x	0.4%	4 of 4
Size	Weekend gaps 1.14x weeknights (since 1993)	1.00x	under 0.1%	3 of 4
Size	Holiday / long-weekend gaps 1.50x weeknights	1.00x	under 0.1%	4 of 4

Size effects say how big moves get, which matters for stops and sizing, not which way to bet. None of these results is tested with trading costs yet.

3b. Tickers tab (one page per ETF)

Where: `output/tickers.html` (overview) and `output/ticker_SPY.html` and so on; code in `tickers/`. The Indexes tab stays the summary. Each ETF gets its own page, grouped as Index ETFs (SPY, QQQ, IWM, DIA, RSP) and 3x ETFs (TQQQ, SQQQ, UPRO, SPXU). A row of ticker buttons under the tab bar switches pages.

Section	What's in it
1. Snapshot	Close; 1 day to 1 year returns; yearly growth since 2010; 52-week range; volatility; worst drop from a peak; dollar volume; moves per 1% SPY move (all, last year, VIX over 25). Price chart for 12 months.
2. Order flow (daily stand-ins)	Volume vs. its 20-day average and how rare that is; where it closed in the day's range; share of volume on up days; the accumulation/distribution line. Related: Kalshi perps for that market, and DIX/GEX on the SPY page.
3. Overnight vs. daytime	Share of growth from overnight, the average night and day, years overnight won, and a growth-of-$1 chart

Section	What's in it
4. After each kind of SPY day	This ticker's next day, next overnight, next daytime and next 5 days after each SPY day type
5. Event days	Gap and daytime sizes on jobs, CPI, Fed, expiration and month-end days
6. How far it can go against you	Bad-day sizes by VIX, a stop table (1% to 10%), opening gaps, and a drawdown chart
7. 3x tracking (3x funds only)	Gap from the exact multiple over 1 day to 3 months, and choppy months
News and calendar	The ticker's Yahoo headlines, Fed press releases, and the next 45 days of scheduled events

- **Overview page:** a side-by-side table of all 9 (close, 1 day to 1 year, volatility, below peak, moves vs SPY), the latest headlines across tickers, Fed releases, and the calendar.
- **Example from Nov 5:** SPY's volume was 3.06x its 20-day average, the 14th heaviest of 4,216 days since 2010, and it closed near the day's low. SQQQ is 99.9995% below its 2010 peak, which shows how inverse 3x funds decay over years.
- **Limits:** per-ticker numbers are descriptive (the luck-tested versions are on the Indexes tab). Order flow means daily stand-ins, because trade-by-trade flow isn't free. News is headlines and links only, and Yahoo's list can include promotional pieces.

4. Leverage tab (Kalshi perps)

Where: `output/leverage.html`; code in `leverage/`. **Data:** Kalshi's public API, with no account needed.

- **A sortable table of all 88 perps (86 active)**, highest leverage first. Filter by type (Index, ETF, Stock, Metal, Energy, Crypto) and platform. Columns: max leverage, move to liquidation, funding, days longs paid, funding payments so far, spread, volume, open interest, hours, and launch date.
- **Four studies on the index perps:** liquidation odds, funding cost, tracking, and the perp's overnight move vs. the ETF's open.

Perp	Max leverage	Move to liquidation (est.)	Long liquidated within 1 day	Within 1 week
US500	12.2x	1.9%	8.2% of days (SPY since 1993)	31.1%
US Tech 100	10.6x	2.2%	13.4% (QQQ since 1999)	38.8%
SPY perp	5.0x	4.6%	0.6%	6.7%
QQQ perp	5.0x	4.6%	2.3%	14.7%

- **Funding:** US500 longs paid about $1.36 a day per $10,000 held (about 5% a year). At max leverage that's about 61% a year of the cash you put up.
- **Tracking:** US500's 4 pm moves lined up with SPY's at 1.00, with an average gap of 0.04% a day (22 days).
- **Overnight:** US500's 4 pm to 9 am move pointed the same way as SPY's opening gap on 95% of 22 nights.
- **Trust:** prices, funding and hours come straight from Kalshi. Max leverage is Kalshi's own estimate. Move to liquidation is calculated by the app and labeled as an estimate.

5. GEX / DIX tab

Where: `output/gex.html`; code in `gex/`; your copy in `data/manual/DIX.csv`.

Part	What it shows
Refresh bar	Cold / Warm / Hot, the reasons behind it, and two buttons: **Get the file** (opens SqueezeMetrics) and **Update** (copies the file in from Downloads, rebuilds the tab, reloads)

Part	What it shows
Today's reading	Latest GEX ($1.84B on Nov 5, above zero) and DIX (37.9%, its lowest in the past year), plus the Indexes tab's VIX (28.2, expected move about 1.78%). Also the Leverage tab's cushions: US500's 1.9% is about 1.8 typical day-session moves on a day like this.
Last 6 months	GEX and the S&P 500, as two charts
History	When GEX closed below zero, the next day session moved 1.31x bigger even after adjusting for VIX (possible edge, size). DIX: no edge either way.
Data check	Your copy's dates, gaps, and the other tabs' data dates

What makes it Cold, Warm or Hot

Worked out from what the other tabs already pulled, each time run_all.py, run_stage1.py or Update runs. It's a reminder of when GEX is worth checking, not a market call.

Level	Rule	Why (history since 2011)
Hot	Any one of these: VIX closed at 20 or higher; your last GEX was below zero; your last GEX was near zero ($0 to $1B); SPY's latest day was a big drop or big rise; the fear curve is inverted (VIX above VIX3M)	GEX was below zero on 17% of days with VIX 20-25, 39% with VIX 25-30, and 46% with VIX 30+. Below zero stayed below zero the next day 70% of the time. Near zero turned negative the next day 14% of the time (vs 3% from higher).
Warm	None of the Hot triggers, but VIX is between 16 and 20	GEX was below zero on 3.9% of those days
Cold	Everything else: VIX under 16, GEX comfortably positive, no big day, normal fear curve	GEX was below zero on about 1 day in 1,800 with VIX under 16
Check mark	Added when your copy runs through the latest market day the app has	Shown as, for example, "Hot: your copy is up to date"

Right now: Hot, with your copy up to date. VIX closed at 28.2, SPY had a big drop (-4.4%), and the fear curve is inverted.

Why the download is manual: SqueezeMetrics' terms bar systematically retrieving their data to build a database. So the app never fetches it; you click their CSV link, and Update only copies the file from your own Downloads folder. It checks the columns, only replaces your copy with something newer, and keeps the old copy as DIX_previous.csv.

6. Where the data comes from, and how far to trust it

Source	Used for	How it's fetched	Trust
Yahoo Finance (yfinance)	ETF prices	Automatic	Free and unofficial. Old opening prices are the weakest part; unconfirmed opens are flagged and left out.
Cboe	VIX, VIX9D, VIX3M, VVIX	Automatic	Official public files
FRED	2-year and 10-year yields	Automatic	Official; shifted a day because they're published late
BLS / Federal Reserve	Jobs, CPI and Fed dates	Typed in by hand	Official calendars; BLS sometimes moves dates
Kalshi public API	Perps: prices, funding, leverage, hours	Automatic	Straight from Kalshi. Leverage is their estimate; move to liquidation is the app's calculation.
SqueezeMetrics	GEX and DIX	You download it	Proprietary method; can't be checked independently
Yahoo Finance headlines	Tickers tab news	Automatic	Unofficial (same as prices); headlines and links only
Federal Reserve RSS	Fed press releases	Automatic	Official feed, published for feed readers

7. How it's checked

- **Self-check (run_stage1.py --check), all passing:** no look-ahead in the daily or long-history tables (both rebuilt from data cut off at 6 dates); overnight x daytime = full day; day-type labels; known-at labels; one expiration per month.
- **A data check on every tab.** Warnings also appear as a banner at the top of the page.
- **Luck tests** have fixed random draws per test, a version for overlapping 5-day windows, and a flag on borderline results.
- **The helper** refuses requests that don't come from the hub's own pages (tested). The Update flow was tested end to end with a test file, and your real copy was restored exactly.

8. Limits, and what isn't built yet

- **No trading costs anywhere yet:** no spreads, slippage, fees or funding in the tested results. Every lead still has to survive them.
- **Daily data only:** no premarket, opening range or VWAP. Order flow is daily stand-ins only.
- **Per-ticker numbers aren't luck-tested** (only the Indexes tab's SPY-based results are).
- **No daily up/down call yet**, and no trade log.
- **Kalshi perps are days to weeks old**, so their own numbers are first looks.
- **The folder belongs to this session** and is deleted if the session is deleted.

9. Stage 2 (next)

- **A daily call:** up, down, or no edge, for 1-day and 1-week holds, built from the Indexes tab's leads, with what would confirm or cancel it.
- **Turning the call into a Kalshi position:** which perp to use, a stop from the stop table well inside liquidation, a size that survives a 1-in-100 day, and extra care before weekends, holidays, jobs and Fed days, or when GEX is below zero.
- **Cost-tested versions of the leads**, plus minute data for entry timing.
- **A log of every call**, so they can be scored.
- **A Refresh-everything button** in the helper.

Worth deciding before stage 2

- Which holding periods matter most to you: overnight, the trading day, or about a week.
- The most leverage you'd consider, which sets the stop and size rules.
- Which perps to cover first: US500 only, or also US Tech 100 and the SPY / QQQ perps.